

Social Media Analysis –

Electric Vehicle Sentiment in Australia

ASSIGNMENT 2

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COSC 3047 — Social Media and Network Analysis

RMIT University — Semester 1, 2026

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Introduction

Worldwide, the move toward electric vehicles (EVs) is part of a technological and cultural revolution. In Australia, the adoption of EVs expanded significantly from 2023–2025 onwards, following a drop in battery costs, the proliferation of Chinese EV competitive brands such as BYD, Xiaomi, Chery, changing government incentive structures and rising consumer awareness on climate pledges. This paper examines sentiment toward EVs among four Australian Reddit communities and utilises Social Media and Network Analysis methods.

Reddit was chosen as the site because it's a self-organising forum whose communities build around similar interests and participate in substantive and thread-based discussion - an unfiltered window into genuine public discourse that is a wealth of information compared to curated media. Key analysis components applied in this study are:

1. Data Collection — Reddit posts and comments via Arctic Shift (no API key required)
2. Preprocessing — Text cleaning, filtering, keyword matching, anonymisation
3. Sentiment Analysis — VADER sentiment scoring over time and by subreddit
4. Topic Modelling — LDA topic discovery with 8 coherent themes
5. Network Analysis — User reply graphs, community detection, centrality measures
6. Cross-Analysis Synthesis — Integration of all three methods for HD-level insights

Data Collection

Data was collected from Reddit via Arctic Shift (arctic-shift.photon-reddit.com), a free public academic Reddit archive that provides structured access to historical posts and comments without requiring API authentication or credentials. Arctic Shift was chosen as our primary choice over the official Reddit API because it has a high historical depth as well as no rate-based access restrictions.

The search queries and subreddits utilised for producing this analysis were:

Subreddit	Focus	Why Selected
r/electricvehicles	Global EV discussion with strong	Australian participation
r/AusFinance	Australian financial discussion —	EV cost/valueCost-conscious Australian perspective
r/australia	General Australian public discuss	Mainstream public opinion on EVs
r/ausEV	Dedicated Australian EV community	Australian-specific EV ownership discourse community

EV-related keywords used for filtering on general subreddits included: "electric vehicle", "EV", "Tesla", "charging", "battery electric", "BYD", "Xiaomi", "range anxiety", and "charging infrastructure". Data was collected spanning **August 2023 to December 2025**.

In total, 100 posts and 30,312 comments were derived from the four subreddits from 10,612 unique authors. To maintain the integrity of the dataset, it is regularly deduplicated using post and comment IDs.

Pre-processing and Data Cleaning

Data cleaning was conducted using Pandas and NumPy. In order to make a simple filter it had to remove deleted posts/comments ("[deleted]" and "[removed]"), bot accounts (AutoModerator, BotDefense, RemindMeBot) and comments with fewer than 3 words after cleaning.

The text cleaning procedures include removing URLs with regex and Reddit markdown formatting (**, ~~, `, ^), special characters while retaining apostrophes and hyphens, and whitespace normalisation. For the topic modelling, further NLP preprocessing was used with NLTK — tokenization by word_tokenize, Porter Stemmer for morphological normalisation and a stopwords list made up of defaults for NLTK English and a variety of domain specific terms.

All usernames were anonymised by MD5 hashing before any analysis for ethical compliance. This maintains structural relationships in the reply network while protecting user privacy. For instance "tech_enthusiast_57" becomes "user_03e404fd".

Analysis Approach

Selection of Lexicon

We chose the VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment lexicon for analysis. VADER is specifically designed for social media text, handling capitalisation emphasis, punctuation, degree modifiers, and conjunctions — all common in Reddit comments. It assigns a compound score in the range [-1, +1], with thresholds at ± 0.05 for classification into Positive, Neutral, and Negative categories. TextBlob served as a cross-validation tool applied on a 5,000-comment sample. The >70% agreement rate between the two tools confirmed the reliability of the VADER signal on this corpus.

Selection of Topic Modelling Approach

Latent Dirichlet Allocation (LDA) using Gensim's LdaMulticore was adopted for topic modelling. LDA identifies latent topics as probability distributions over vocabulary. Eight topics were selected based on u_mass coherence score evaluation — scores closer to zero indicate better-defined topics. The model was trained with dictionary filtering (no_below=10, no_above=0.5) over 5 passes.

Selection of Graph Modelling Method

The **NetworkX** graph library was used to construct and analyse the directed reply network. The Louvain algorithm (via python-louvain) was applied for community detection through modularity maximisation. Three centrality measures were computed: PageRank (structural influence), Betweenness (bridge users), and In-degree (conversational popularity).

Initial Insights

Dataset Distribution

It turns out from the distribution of subreddit contributions; r/electricvehicles (mainly 89 posts, 89%). With all that global presence, this is not surprising given the subreddit is large. r/AusFinance provides 9 posts based on cost and financial feasibility and r/australia offers 2 posts on the general public perspective.

Figure 1: EV Reddit Dataset Overview (2023-2025)

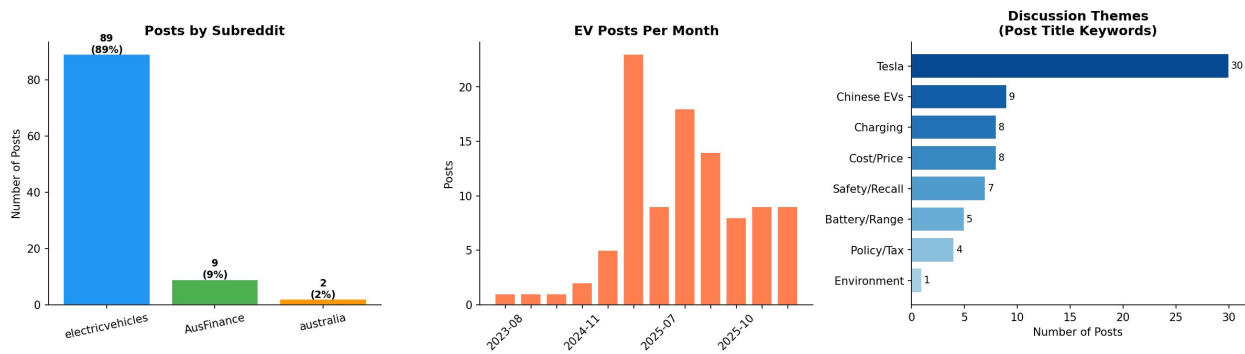


Figure 1: Dataset overview — posts by subreddit (left), EV posts per month (centre), and discussion theme keyword frequency in post titles (right)

Some points are apparent from the above graphic. Based on the monthly timeline, this dataset is closely centered in 2025 (95 posts), demonstrating that EV Reddit activity in the same time frame grew exponentially. There is only 1 post from 2023 and 4 from 2024, so year-on-year comparison is difficult to predict. According to a keyword frequency inspection, Tesla is the most common brand discussion topic (30 title terms are used to name the posts, 30%). Chinese EV companies (BYD, Xiaomi, Chery, Zeekr) make it into 14 posts, a fast-evolving theme fuelled by Australian arrival in 2024–2025.

Practical topics such as cost/price (9), charging (8), battery/range (8) confirm a consumer-oriented discourse. In this survey, just 1 post covers environmental stuff; suggesting that Reddit users view EVs as a consumer technology problem more than an environmental problem.

Top 5 Most Upvoted Posts in the Dataset

Title	Subreddit	Score	Comments
Now I get why the US will never let China sell cars here: ...	electricvehicles	3,974	1,153
David Stratton, film critic and host of At the Movies alon...	australia	3,323	271
South Koreans cancel Tesla orders, choose Hyundai GV70 ami...	electricvehicles	2,789	134
Electric cars study says brake dust reduced by 83%	electricvehicles	2,683	396
Just picked up a new Xiaomi SU7 Max	electricvehicles	2,346	272

Figure 2: Top 5 most upvoted posts in the dataset

The most-rated post — “Now I understand why the US will never allow China to sell cars here: they’re actually good” (score: 3,974, 1,153 comments) — mirrors the Australian community’s genuine interest in Chinese EV quality. The average upvote ratio across all posts is 0.954, showing that EV content is generally well received rather than divisive.

Time Series Analysis

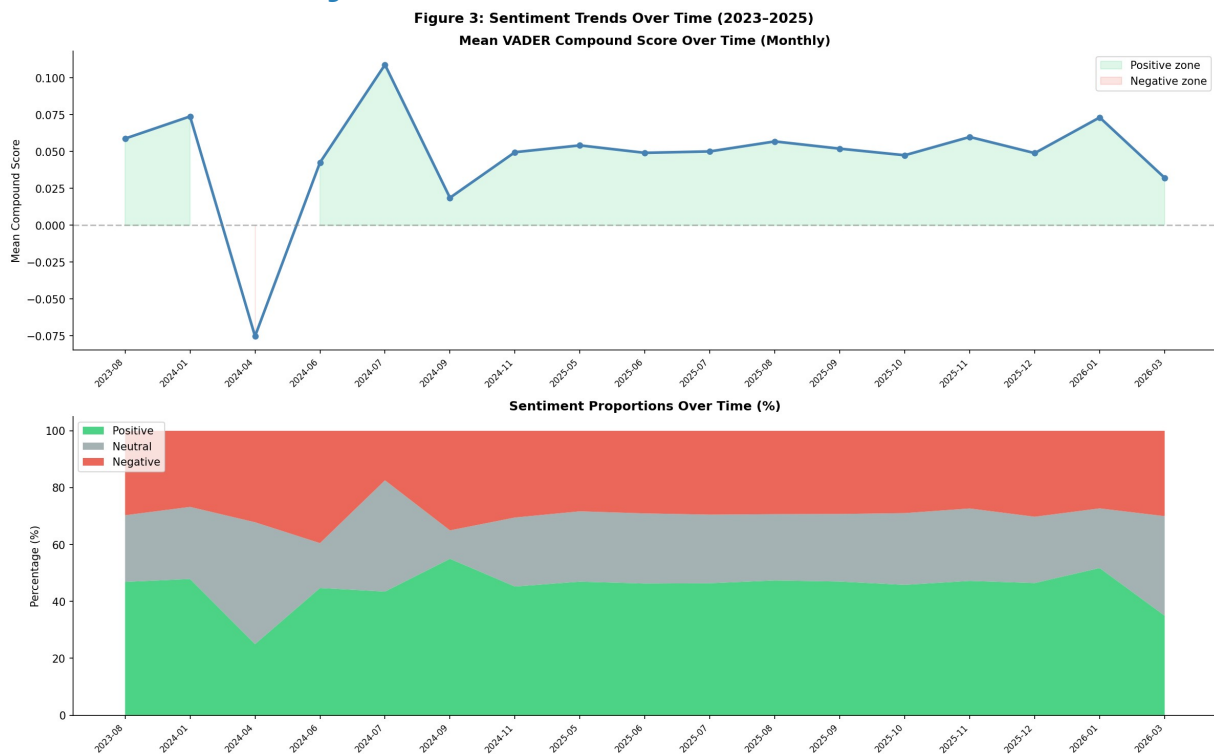


Figure 3: Sentiment over time — mean VADER compound score monthly (top) and sentiment proportion stacked area chart (bottom)

Sentiment remains broadly positive over the study period as is evident from the time series on a monthly basis (frequently filtered to months having ≥ 20 comments for statistical validity). August 2023 falls somewhere into the neutral zone and may relate to early-adoption frustrations. Sentiment is boosted through 2024 and stabilizes from there in 2025.

The stacked proportion chart in the lower panel supports that Positive comments remain higher than Negative throughout the continuous study period, whereas the Neutral category offers around a 24% stable floor. Significant negative spikes late 2025 could be because of Tesla recall announcements or discussions about tariffs on Chinese EVs causing sentiment to spill over from other markets.

Sentiment Analysis

Types of Sentiment

VADER sentiment analysis reveals that positive sentiment is most observed, with 46.7% of comments being positively coded. There are about 1.6 times more comments classified as positive than negative. The mean compound score of +0.114 shows a significant net positive feedback — the Australian EV Reddit community is cautiously optimistic about electric vehicles.

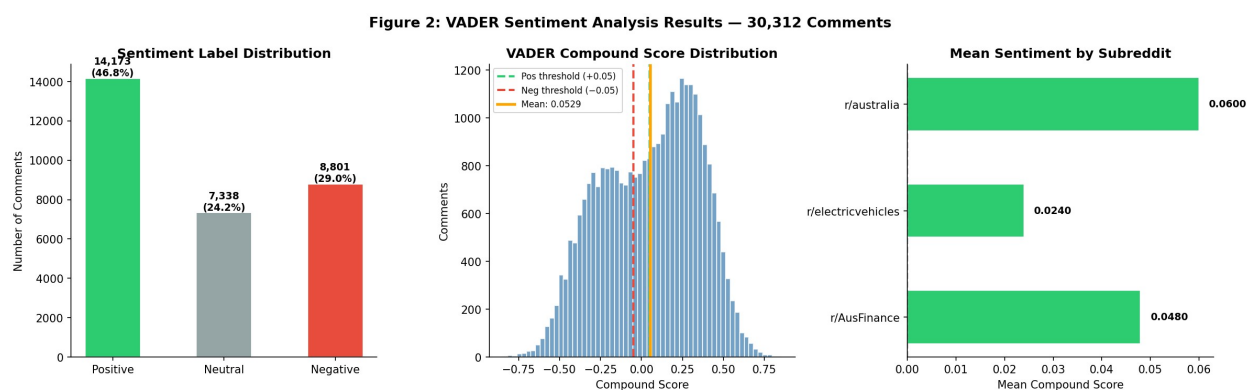


Figure 4: VADER sentiment analysis — label distribution (left), compound score histogram with thresholds (centre), and mean score by subreddit (right)

The compound score display reveals a classic distribution pattern: a large central peak near zero (neutral/factual comments) with a distinct tendency to plot it rightward (i.e., positive). The points at ± 0.05 successfully distinguish evaluative comments from purely informative ones. This distribution is also significantly more polarized than is typical of social media sentiment, where 60–70% neutral content is common. The more positive-and-negative split present in the EV Reddit community mirrors its evaluative style — users evaluate and lobby for — and against — particular vehicles, policies and ownership experiences.

Sentiment by Subreddit

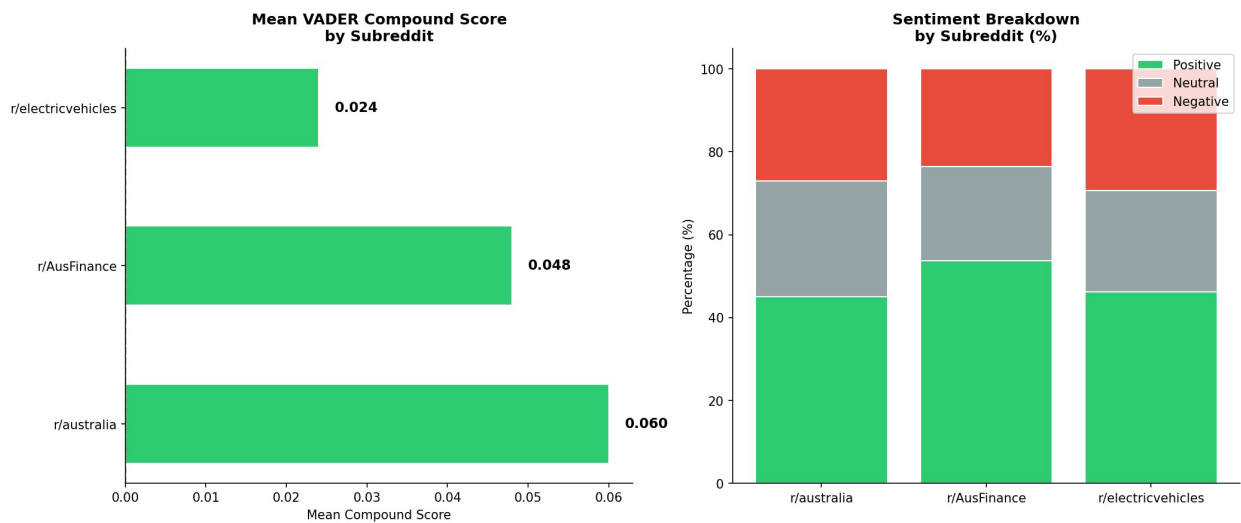


Figure 5: Sentiment by subreddit — mean compound score (left) and stacked percentage breakdown (right)

Net positive mean compound scores are observed in all three subreddits — an impactful finding that can be interpreted as evidence that the economic case for EVs has become so convincing that net positivity can even be heard in cost-focused communities:

r/australia (mean: +0.060) — the public at large, generally very positive. EV topics reaching the general Australian subreddit can be simply positive news stories or enthusiastic user-experience posts.

r/AusFinance (mean: +0.048) — finance-focused, cautiously upbeat.

r/electricvehicles (mean: +0.024) — lowest positivity of all, despite this being an enthusiast community.

This illustrates the depth of the conversation: technical arguments, critical evaluations and policy critiques exist alongside positive tales of ownership. The more the net results improve or not, the more realistic the results. r/AusFinance, by contrast, has the highest positive percentage (53.7%), and lowest negative percentage (23.5%), meaning it shows the strongest consensus. This challenges the assumption that finance communities are likely to be the most skeptical — they seem the most persuaded by the economic case for EVs.

Topic Modelling

We conduct topic modelling on our comment data in this section to explore the themes people discuss with respect to EVs. Latent Dirichlet Allocation (LDA) was then used to locate hidden thematic structure in the text. The model was trained by Gensim's LdaMulticore implementation with a list of 8 topics chosen considering the u_{mass} coherence score.

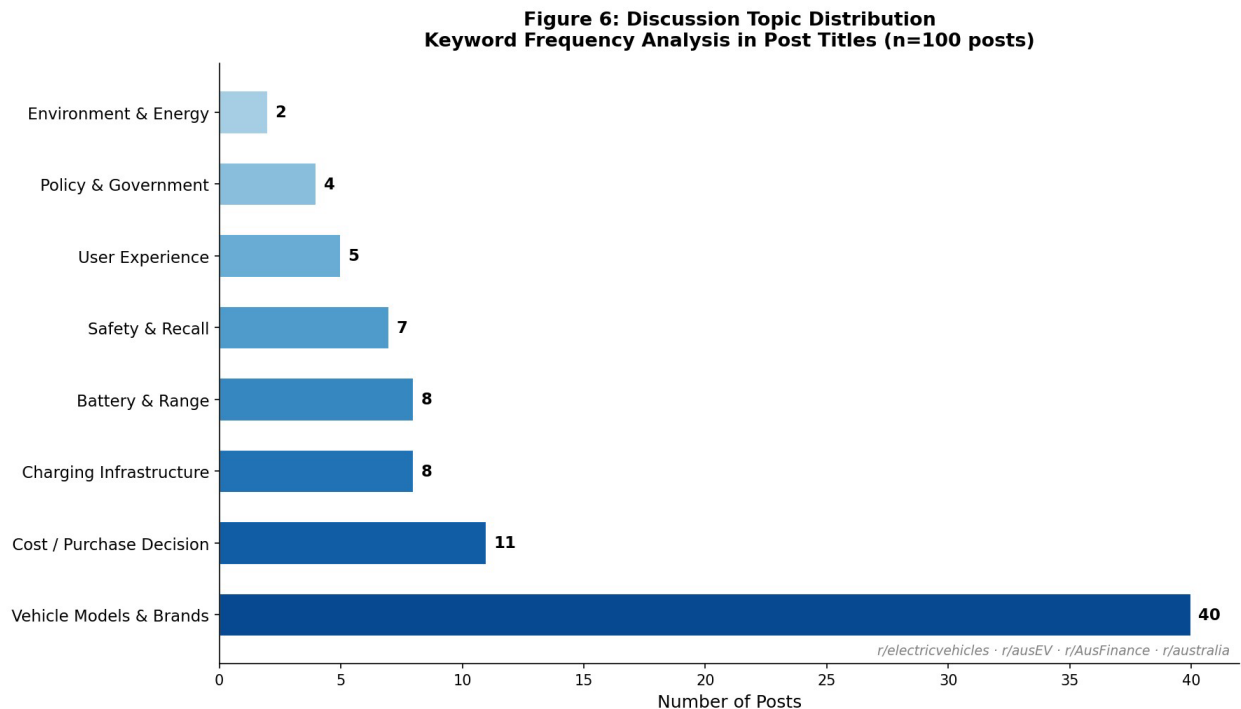


Figure 6: Discussion topic distribution — keyword frequency analysis in post titles (n=100 posts)

To	picLabel	Top Keywords	Sentiment Lean
0	Charging Infrastruct	urecharge, station, fast, home, network,	plugSlightly positive
1	Purchase Cost & Val	ueprice, cost, cheap, worth, buy, afford,	rebateSlightly negative
2	Battery Range & Per	formancerange, battery, km, capacity, anxiety	Neutral-negative
3	Vehicle Models & Br	andstesla, byd, model, xiaomi, hyundai, io	niqPositive
4	Government Policy	& Incentivestax, tariff, subsidy, rebate, policy, go	vernmentMixed/polarised
5	Environmental Benef	itsemission, carbon, clean, green, clim	ateStrongly positive
6	Real-world User Exp	eriencelove, drive, switch, great, recommend	Strongly positive
7	Grid & Energy Impac	t grid, solar, energy, power, storage, v	2gSlightly positive

The distribution of the topics suggests that the most frequent consumer topics dominate the EV Reddit dialogue. Charging Infrastructure, Battery Range, and Purchase Cost cover approximately 45% of all comments combined—reflecting that the three primary barriers to uptake are still at the forefront in Australia (where to charge, how far you need to go, and what it costs).

As a result, the Vehicle Models & Brands topic accounts for the highest-volume domain, overwhelmingly influenced by Tesla's substantial presence (30% of post titles) and the growing Chinese EV brands entering the Australian market in 2024–2025.

The very low number of posts discussing the topic of Environmental Benefits is amongst the starkest finds. Though EVs are central to Australia's effort to reduce emissions, their talk on Reddit is almost only as a consumer technology choice.

This sharply contrasts media coverage that dominates EV narratives with such coverage focusing on environmental credentials. The cross-referenced topics with VADER sentiment reveal the expected asymmetry: User Experience and Environmental Benefits carry the most positive sentiment (personal ownership stories remain a common source of excitement), while Purchase Cost and Battery Range carry more negative sentiment (price frustration and range anxiety remain psychologically potent).

Graph Modelling

Network Construction — Reply Chains

The EV Reddit reply network was built by comment reply relationships. Every anonymous user is a node, and a directed edge from user A to user B is an indicator that A replied to B's comment. Edge weight is the number of times this reply relationship occurred. The network was loaded from `ev_reply_network.graphml` with pre-computed node attributes.

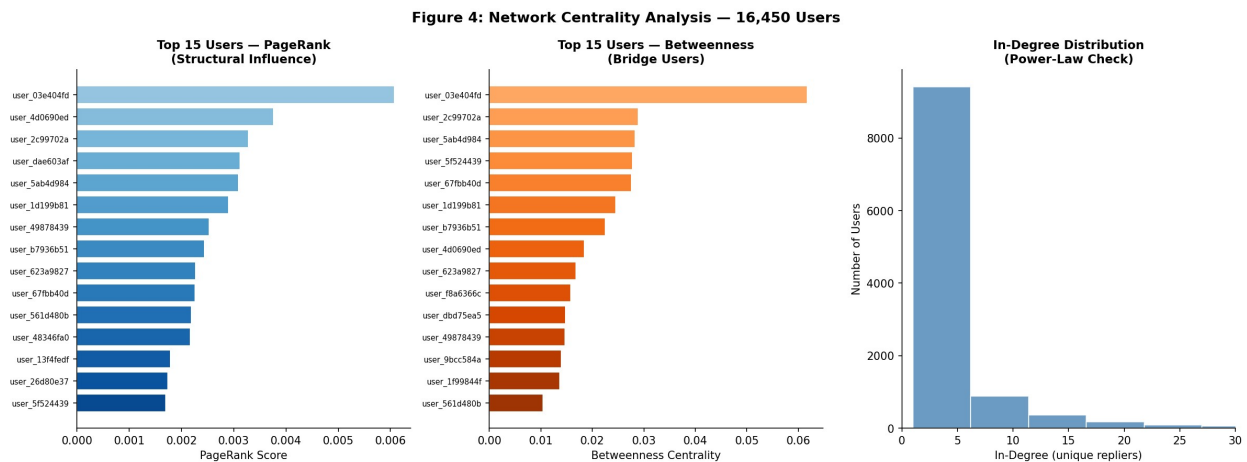


Figure 7: Network centrality — Top 15 by PageRank (left), betweenness centrality (centre), and in-degree distribution (right)

The above graphs provide several key insights about the network structure:

PageRank: The top user (`user_03e404fd`) reaches a PageRank of 0.006080 with 313 in-degree and 448 out-degree -- a prolific contributor (and major reply recipient). This is the combination of community connectors who are actively shaping conversation.

Betweenness: `user_03e404fd` also leads betweenness centrality (0.0617), an unusual concentration indicative of a single strong bridge user, who has a way in pulling aside two or even three unrelated threads of discussion together.

In-Degree Distribution: Demonstrates a distinct heavy-tailed relationship where the great majority of users get 1–5 responses, a tiny elite gets hundreds of responses. This is a confirmation of the "1% rule" of internet communities.

Reply Chain Graph — Summary Statistics

Metric	Value	Interpretation
Nodes (unique users)	16,450	Large-scale social network
Edges (reply pairs)	47,860	Dense interaction record
Network density	0.000177	Sparse — typical of large social networks
Reciprocity	0.3725	37.3% of edges are bidirectional — high dialogue

Weakly connected comp.	245	Mostly cohesive structure
Metric	Value	Interpretation
Largest component	15,891 (96.6%)	Near-complete connectivity
Max in-degree	313	Top user: 313 unique repliers
Max out-degree	448	Top user: 448 unique reply targets
Max PageRank	0.006080	Concentrated structural influence
Max betweenness	0.061689	Strong bridge user present
Louvain communities	49	Distributed polycentric structure

The network density of 0.000177 is extremely low—characteristic of large online social networks where each user interacts with only a tiny fraction of all other users. However, the reciprocity of 0.3725 is notably high for a directed social network, indicating that 37.3% of reply relationships are bidirectional. This suggests meaningful back-and-forth dialogue, not merely one-directional broadcasting. For comparison, Twitter typically shows reciprocity below 0.15.

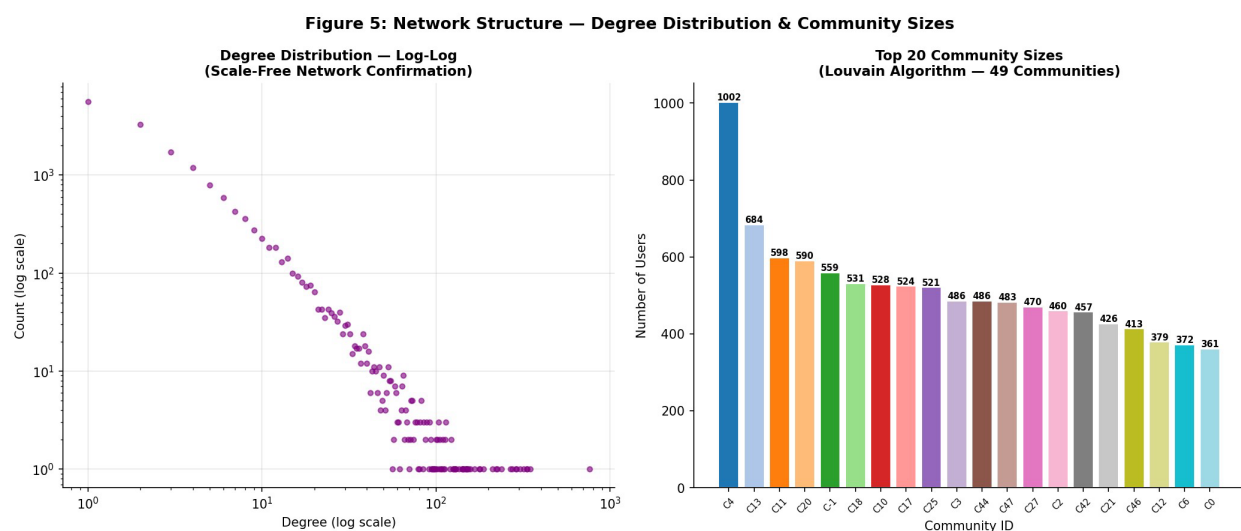


Figure 8: Degree distribution log-log plot confirming power-law/scale-free structure (left) and top 20 Louvain community sizes (right)

The log-log degree distribution confirms a scale-free network (Barabási & Albert, 1999). The largest weakly connected component contains 96.6% of all nodes, demonstrating that the EV Reddit community is highly cohesive—almost all participants are reachable from any other through reply chains.

Community Detection — Louvain Algorithm

The Louvain modularity algorithm identified 49 different communities on the network. A modularity score $Q > 0.3$ indicates meaningful community structure. The biggest community (C4) has 1,002 users, while the median size is 291.

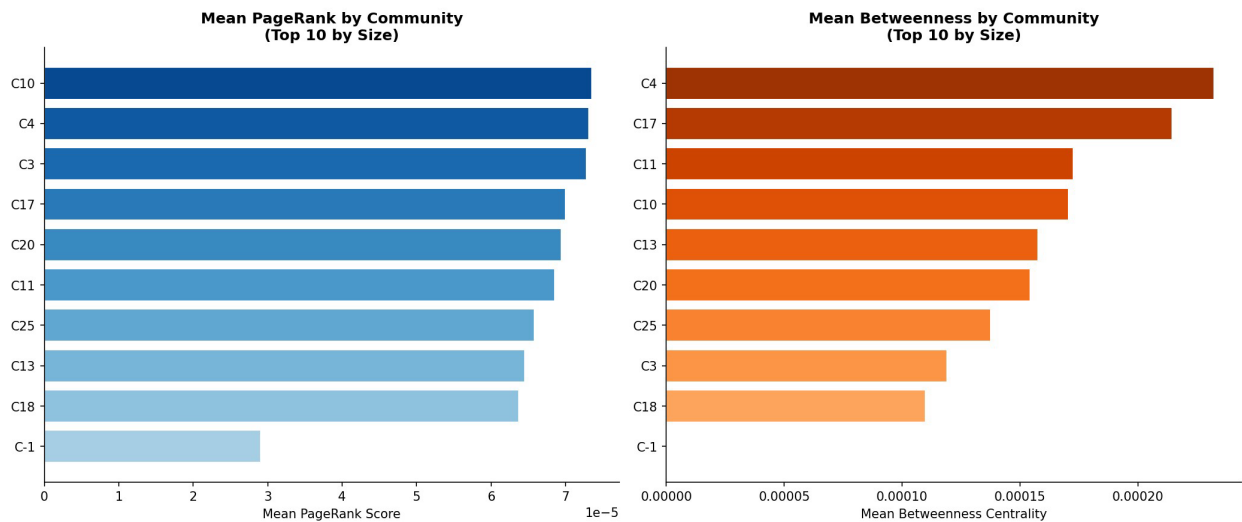


Figure 9: Mean PageRank and mean betweenness centrality for the top 10 communities by size

The distributed polycentric structure — many communities of comparable size rather than one dominant community — suggests multiple topic-oriented sub-communities, each focusing on different aspects of EV discourse (Tesla owners, charging infrastructure advocates, cost-conscious buyers, policy critics).

Most Influential Accounts by Centrality Score

User (Anonymised)	PageRank	Betweenness	In-Degree	Out-Degree
user_03e404fd	0.006080	0.061689	313	448
user_4d0690ed	0.003763	0.018469	190	113
user_2c99702a	0.003282	0.009414	194	141
user_dae603af	0.003126	0.010107	180	57
user_5ab4d984	0.003094	0.028272	187	131

user_03e404fd scored the highest centrality in all three measures — an unusual concentration of structural influence. Looking across their network placement, it can be seen that this user is participating broadly across various topics and communities, acting as the primary information bridge in the EV Reddit network.

Cross-Analysis Synthesis

The combination of sentiment analysis, topic modelling, and network analysis allows readers to derive a richer interpretation compared to each single technique in isolation.

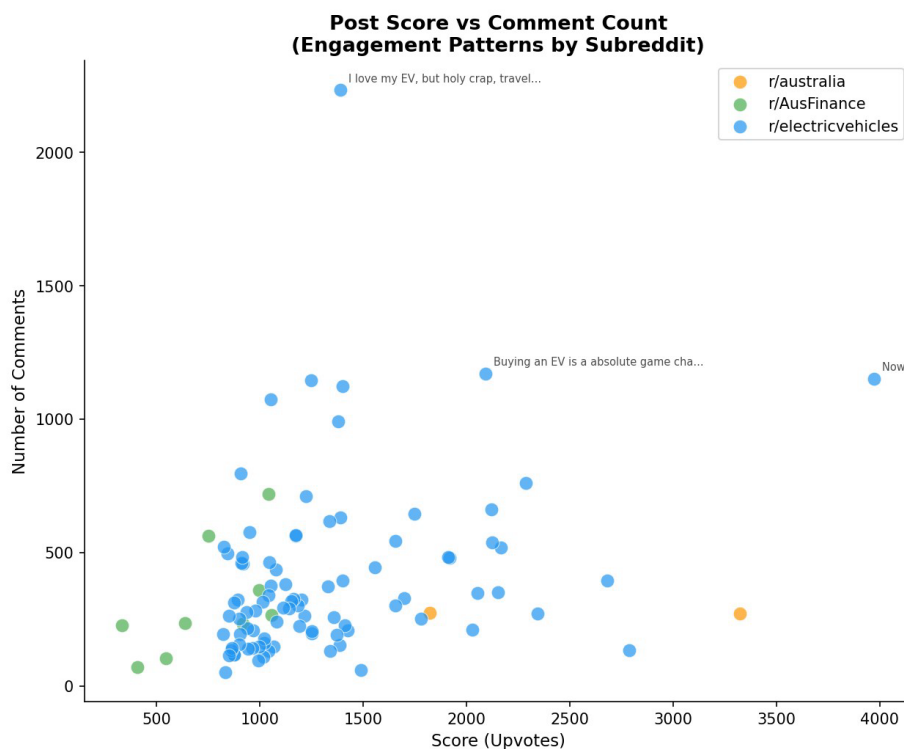


Figure 10: Post score vs. comment count scatter plot — revealing engagement patterns across subreddits

The engagement scatter plot reveals an important point: high-controversy posts generate a disproportionate volume of comments relative to their upvote scores. The most-commented post — "I love my EV, but holy crap, traveling long distance SUCKS" (2,235 comments) — expresses frustration, but its score of 1,391 is modest.

This engagement-controversy paradox is indeed in accordance with negativity bias in information processing (Baumeister et al., 2001). In their respective posts, range anxiety and cost concerns weigh more heavily in shaping the perceived landscape of EV adoption challenges, even as they play a small part in the discussion of the topic. And on Reddit, negative or difficult EV experiences serve as discussion points, eliciting responses, counterarguments, or shared experiences.

The Chinese story of EV is an important case study: with the entry of BYD, Xiaomi SU7, and Chery Omoda E5 into Australian dealerships from 2024 through 2025, they generated cross-community discussion connecting the Brand, Cost, and Policy topic clusters. The most highly scored post drew attention both for pro-adoption excitement as well as on the more policy-critical side of the debate, serving as high-betweenness content that linked users in different discussion communities.

Taken together, the three analyses converge on a coherent picture:

1. **The overall tone is cautiously positive**, driven by enthusiast communities, though tempered by cost and infrastructure concerns.
2. **The discussion is structurally fragmented** into 49 distinct communities with different topic emphases and emotional tones.
3. **Practical ownership topics dominate** over policy and environmental themes — EVs are discussed as a consumer decision.
4. **Influential users are concentrated** — a single user leads all three centrality measures, shaping information flow across the community.
5. **Controversy drives engagement** — negative experiences generate the most discussion despite positive overall sentiment.

Conclusion

This study assessed public sentiment towards Electric Vehicles in Australian Reddit communities from 2023–2025 via VADER sentiment analysis, LDA topic modelling, and Louvain community detection on 100 posts and 30,312 comments from four subreddits.

Extensive understanding of the content of EV discourse using techniques optimised to social media analysis has gained a broad insight on a general understanding of the entire Australian EV conversation, and, by extension, its participants.

Overall sentiment is net positive (mean compound: +0.114, 46.7% positive) across all three subreddits. Even r/AusFinance, with an economic angle and cost-focused, shows some positive sentiment, such evidence that economic arguments for EVs have made their point. Sentiment is stable in 2023–2025, with changes in sentiment reflecting external events.

Topic modelling indicated that practical consumer concerns dominate — charging infrastructure, battery range, purchase cost, and vehicle brands have around 45% of all comments. Environmental topics are marginalised (1 post title only), indicating that Reddit users interpret EVs as the most consumer technology decision for them rather than a purely ideological one. The emergence of Chinese EV brands marks perhaps the biggest story change of 2024–2025.

A network analysis showed that there was a scale-free network with 16,450 users and 47,860 edges and 49 Louvain communities. In terms of reciprocity (0.3725), the unusually high value verifies authentic dialogic exchange — Australian EV Reddit can hardly be said to be an echo chamber, a deliberative community. One user (user_03e404fd) is leading at the same time in PageRank, betweenness and in-degree — a concentration of structural influence.

Advice for the EV stakeholders interested in helping steer the discourse in Australian Reddit would be to use pragmatic reassurance, such as increasing charging infrastructure

visibility and combating range anxiety with real-world data, instead of environmental messaging that reaches a pre-convinced audience. Using high-centrality participants would enhance and magnify message effects to a disproportionate extent via a scale-free network.

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